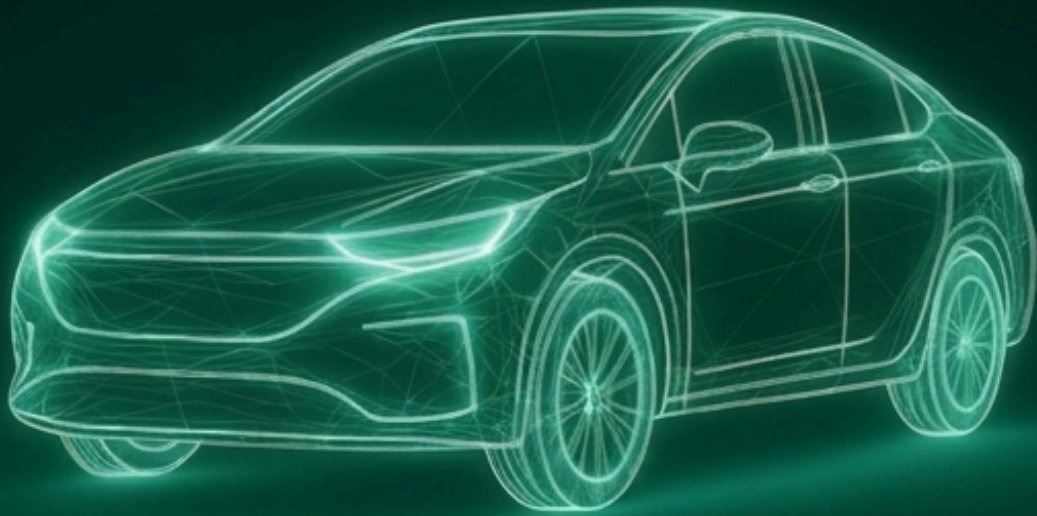


Electrical Vehicles & Renewable Energy Production



State Level Data Analytics Capstone

Executive Summary

This analysis investigates a widely assumed relationship: that states with cleaner electricity grids adopt electric vehicles (EVs) at higher rates. Using 2018 vehicle registration data and 1990–2018 energy generation records across all 50 US states and Washington, D.C., this study tests that assumption statistically rather than relying on visual impression.

The result challenges the common narrative. Renewable energy share shows only a weak and statistically insignificant correlation with EV adoption ($r = 0.24$, $p = 0.088$). States with the cleanest grids — such as Vermont at 99.8% renewable — rank among the lowest in EV adoption, while California, with a moderate 43% renewable share, leads the nation by a wide margin.

The data instead points to policy — not grid composition — as the more likely driver of EV adoption. States leading in EV registrations are consistently the same states with direct purchase incentives or Zero- Emission Vehicle (ZEV) mandates, regardless of how clean their electricity supply is.

Total EVs Registered (2018)	543,610
States & Territories Covered	50 + DC
Correlation (renewable % vs EV adoption)	$r = 0.24$ (weak)
Statistical Significance	$p = 0.088$ (not significant)
National Renewable Share Growth	9% (1990) → 17% (2018)
Top EV Adoption State	California — 8.28 EVs / 1,000 vehicles

1. Objective & Background

The transition to sustainable transportation is often framed as two connected efforts: cleaning the electricity grid and encouraging electric vehicle adoption. This project examines whether those two efforts are, in practice, statistically linked at the state level, or whether they operate as independent policy tracks.

Guiding Questions

- Does a state's renewable energy share predict its EV adoption rate?
- How has renewable energy production evolved nationally and by state since 1990?
- Which states lead in EV adoption, and what do they have in common?
- What role does policy—rather than energy infrastructure—appear to play?

2. Data Sources & Methodology

Four public datasets were combined for this analysis:

- **EV Registrations (2018):** State-level electric vehicle registration counts.
- **All Vehicle Registrations (2018):** Total motor vehicle registrations per state, used to normalize EV counts into an adoption rate rather than a raw count (which would otherwise favor large states like California by default).
- **Energy Generation by Source (1990–2019):** Annual electricity generation by state, producer type, and energy source, sourced from U.S. energy generation records.
- **State Reference Codes:** Used to standardize state names and codes across all sources.

Data Cleaning Challenges

Government datasets of this kind are rarely analysis-ready. Three notable issues were identified and resolved:

- **Inconsistent state naming:** Washington D.C. appeared under three different labels across the four files (“Dist. of Col.”, “District of Columbia”, “District Of Columbia”), which would silently break joins if not standardized.
- **Producer-type duplication:** The energy dataset reported generation under six overlapping producer categories. Without filtering to “Total Electric Power Industry” only, generation totals would have been double- or triple-counted.
- **Malformed government tables:** The vehicle registration file used a multi-row merged header starting 12 rows into the sheet, with footnote markers embedded in state names, requiring targeted parsing.

After cleaning, all three core datasets matched at 51/51 states (50 states + DC) with zero unresolved nulls.

Key Metric: EV Adoption Rate

Raw EV counts favor largest states by default. To enable fair comparison, EV adoption was calculated as EVs registered per 1,000 total vehicles in each state — this is the primary metric used throughout this report.

3. Finding 1: EV Adoption Is Highly Concentrated

EV adoption is not evenly spread across the country. The distribution is heavily right-skewed — most states cluster at very low adoption rates, with a small number of outlier states pulling far ahead.

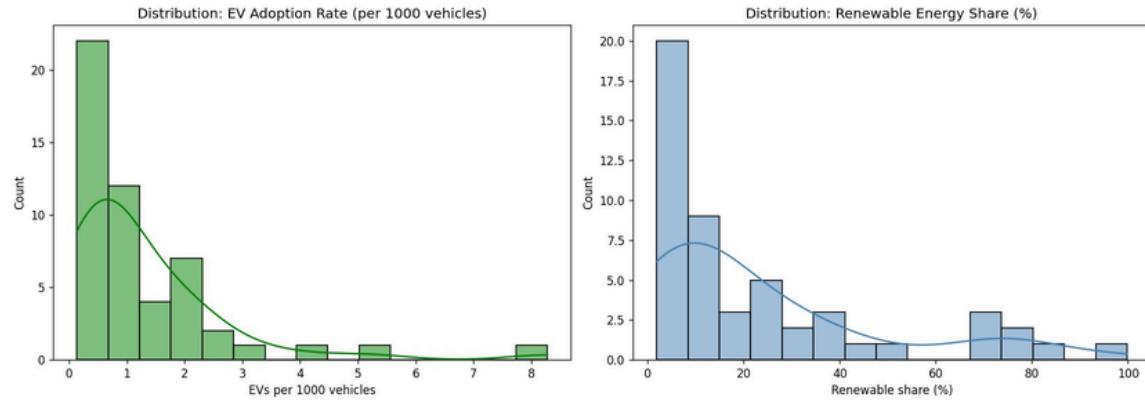


Figure 1: Distribution of EV adoption rate and renewable energy share across states

California stands alone at 8.28 EVs per 1,000 vehicles — more than 1.6x the second-highest state (Hawaii, 5.20) and nearly 2.1x the third (Washington, 3.97). The median state sits at just 0.71 EVs per 1,000 vehicles, underscoring how concentrated national EV adoption still was as of 2018.

4. Finding 2: Renewable Energy Does Not Predict EV Adoption

A Pearson correlation test was run between each state's renewable energy share and its EV adoption rate.

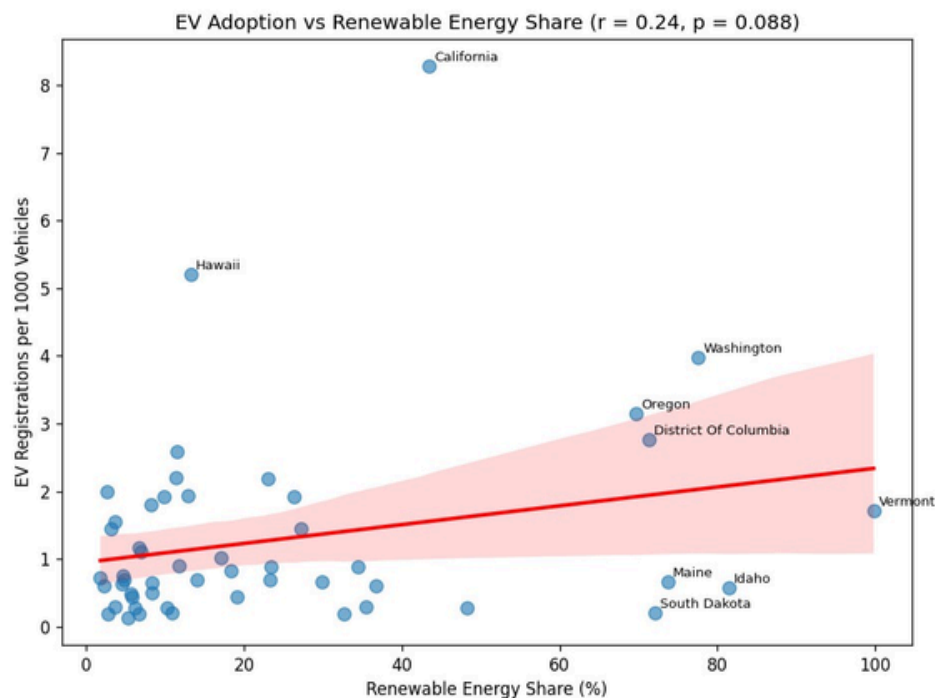


Figure 2: EV adoption rate vs. renewable energy share, with regression line ($r = 0.24$, $p = 0.088$)

Interpreting the Statistics

- **$r = 0.24$ (correlation coefficient):** measures the strength of the relationship on a scale from -1 to +1. A value this close to zero indicates a weak relationship at best.
- **$p = 0.088$ (p-value):** measures the probability that a relationship this strong could occur by random chance alone if no real relationship existed. The conventional threshold for statistical significance is $p < 0.05$. At $p = 0.088$, this result does not clear that bar — meaning the weak positive trend observed cannot be confidently distinguished from coincidence.

The scatter plot makes the disconnect visible: Vermont (99.8% renewable) and Idaho (81%) both show low EV adoption, while California (43% renewable) and Hawaii (13% renewable) rank among the highest adopters. If clean electricity reliably drove EV adoption, these outliers would not exist in this pattern.

5. Finding 3: Renewable Energy Growth Is a Midwest Wind Story

Nationally, renewable energy's share of total electricity generation nearly doubled between 1990 and 2018, rising from approximately 9% to 17%, with the steepest growth occurring after 2015.

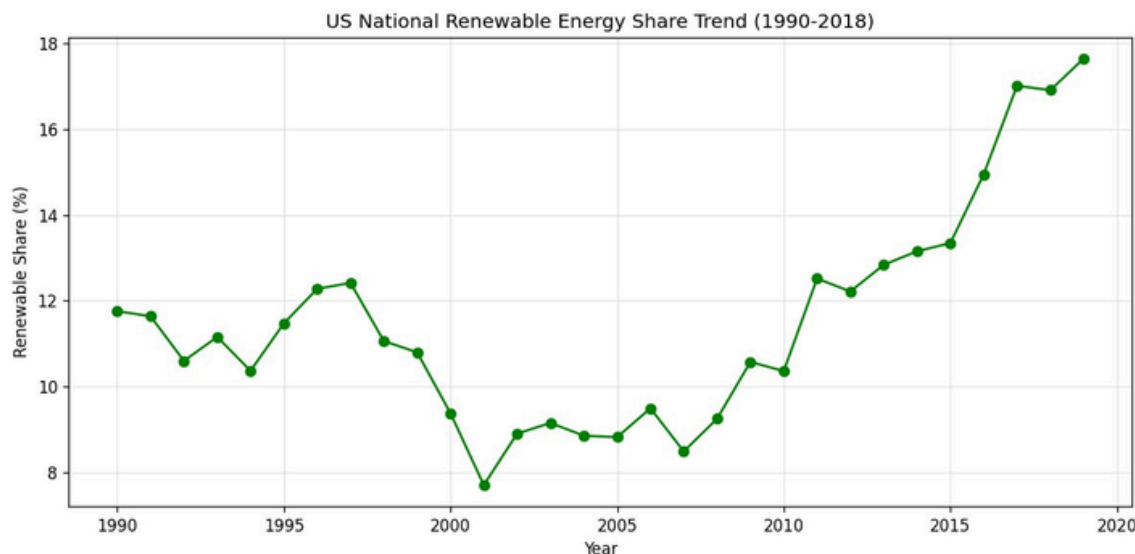


Figure 3: US national renewable energy share, 1990–2018

The states with the largest renewable share growth over this period were not the coastal states typically associated with clean energy branding. Kansas grew from near 0% to 36.8% renewable, Iowa from 3.0% to 35.5%, and Oklahoma from 6.4% to 34.5% — all driven predominantly by wind power expansion in the Great Plains. Notably, none of these states rank among the top EV adoption states, reinforcing Finding 2: renewable energy growth and EV adoption are independent trends.

6. Finding 4: Geographic Patterns

Mapping both metrics side by side shows the states with the highest EV adoption (West Coast, Northeast) do not consistently overlap with the states leading in renewable energy share.



Figure 4: EV adoption rate by state, 2018

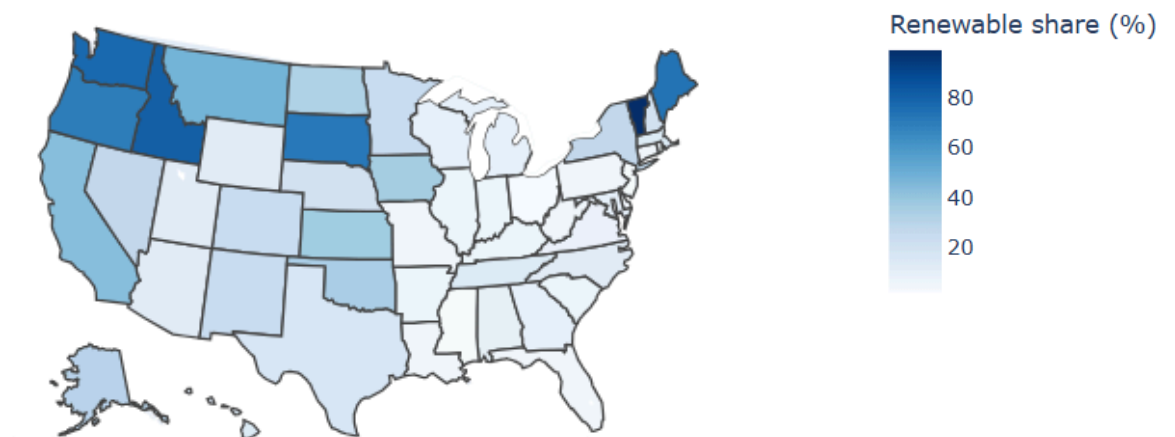


Figure 5: Renewable energy share by state, 2018

Washington and Oregon are the clearest examples of overlap — both rank high on both metrics. But this is the exception, not the rule: most high-EV states (California, Hawaii, New Jersey, Massachusetts) have only moderate renewable shares, while most high-renewable states (Vermont, Idaho, South Dakota) show minimal EV adoption.

7. Policy Context: What Actually Explains EV Adoption

If grid composition doesn't explain the pattern, policy does. Reviewing the leading EV adoption states reveals a common thread: coordinated regulatory and financial incentive programs, not clean electricity.

- **California:** operates a Zero-Emission Vehicle (ZEV) mandate requiring automakers to sell a rising percentage of zero-emission vehicles, paired with the Clean Vehicle Rebate Project offering direct cash rebates to buyers.
- **The multi-state ZEV coalition:** Connecticut, Maine, Maryland, Massachusetts, New Jersey, New York, Oregon, Rhode Island, and Vermont signed a Memorandum of Understanding with California to coordinate EV infrastructure and incentive policy — which explains why several of these states appear clustered together in EV adoption despite having very different renewable energy profiles.
- **Colorado:** offers an independent state tax credit for EV purchases, outside the ZEV coalition.

This distinction matters for policy design: a state cannot assume that renewable energy investment will translate into EV adoption. These are two separate levers requiring two separate policy interventions.

8. Conclusion & Recommendations

Key Findings

- EV adoption is highly concentrated, with California far ahead of every other state.
- Renewable energy share does not reliably predict EV adoption ($r = 0.24$, $p = 0.088$ — not statistically significant).
- National renewable energy generation nearly doubled from 1990 to 2018, driven largely by wind power growth in the Great Plains, not coastal states.
- EV-leading states share common purchase incentive and mandate policies, not common grid composition.

Recommendations

- **For policymakers:** Treat EV adoption and grid decarbonization as separate policy goals. Direct purchase incentives, tax credits, and ZEV-style mandates appear far more predictive of EV adoption than renewable energy investment alone.
- **For industry stakeholders:** Prioritize market entry and infrastructure investment in states with active EV policy frameworks (ZEV coalition states, Colorado), rather than assuming states with clean grids are natural early markets.
- **For researchers:** Future work should incorporate a direct policy dataset (state incentive value, mandate status) to formally test policy as a predictor of EV adoption, rather than inferring it qualitatively as this report does.

Limitations

- EV registration data is available for a single year (2018) only; true time-series modeling of EV adoption over time was not possible with this dataset.
- Policy impact was assessed qualitatively based on publicly documented programs, not a structured policy dataset — this section identifies a likely explanation rather than statistically proving it.
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Correlation analysis establishes association, not causation; unobserved factors such as urban density, charging infrastructure, and regional EV manufacturer presence were not directly measured.